

Kalman filter

Bay's theorem

Predict & correct

Kalman gain: (0) → (1)

mathematical model dependency

sensor dependency

$$X \in N(\mu_A, \sigma^2) \quad \# 1D$$

$$X \in N(\hat{x}, P) \quad \# \text{multi-dimension}$$

State estimation

Covariance matrix

$$\hat{x}_{m,n} = \hat{x}_{m,n-1} + K(z_n - \hat{x}_{m,n-1})$$

System Model (linear ~~FF~~)

motion model

$$X_K = A X_{K-1} + B U_K + W_K$$

State matrix State-time control control matrix

motion uncertainty

$N(0, Q)$

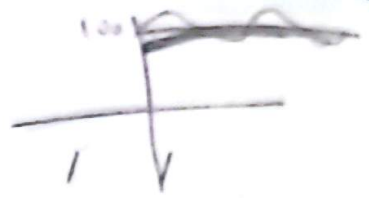
Noise

sensor model

$$Z_K = H X_K + V_K$$

matrix sensor uncertainty

input $H X_K$ + noise $\rightarrow Z_K$

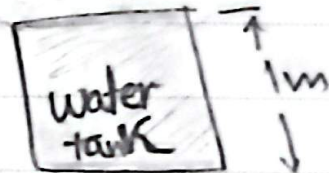


$\hat{x}_{0,0}$
 $P_{0,0}$

$$P = (1 - K_0) * P_0$$

$$K_n = P_{n,m-1} H^T (H P_{n,m-1} + R)^{-1}$$

Problems - ID -



$L = 100\text{cm}$
True

$$\hat{x}_{1,0} = 0$$

$$P_{1,0} = 6^2 + \frac{6^2}{10}$$

$$x_k = x_{k-1} + w_k \quad \#ID$$

Noise = 0.0

$$\sigma^2 = 0.0001$$

$$x_{1,1} = 0 + K(z_1 - 0) \quad z_k = x_k + u_k$$

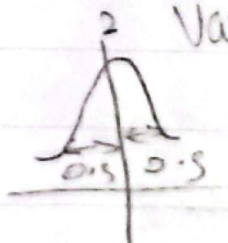
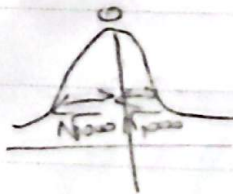
$$x_{2,1} = x_{1,1} + K(z_2 - z_1) = \checkmark$$

$$\rightarrow x_{0,0} = 0$$

$$P_{0,0} = 1000$$

$$\sigma_v = 0.5$$

$$\text{Var} = \sigma^2 v = 0.2 \text{ cm}^2$$



for Same Position

$$\begin{bmatrix} z_1 = 99.17 \\ z_2 = 100.6 \\ z_3 = 100.12 \\ z_4 = 99.61 \end{bmatrix} \rightarrow x_k = \checkmark$$

ALADIB 100

ZOOM